

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/*Final*

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA102: Software Engineering (*New Course*)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Why is it essential to follow software engineering principles during software development? Explain how prototyping technique help in coping with change.
2. What do you mean by software project planning? Explain various cost estimation techniques in detail.
3. What are the deliverables of software design? Explain software design concepts and principles.

Group B

Answer SEVEN questions.

7×8=56

4. What do you mean by requirement elicitation? Explain few traditional as well as modern methods of requirement elicitation.
5. What is software testing? Explain integration and system testing.
6. Why is quality a mandatory parameter in software? Explain the various activities performed during quality assurance in software.
7. What is software dependability? Explain the concept of software reuse.
8. Explain the concept of cloud computing and Big data. What type of challenges are addressed by these emerging technologies?
9. What is agile software development? How does this methodology mitigate the issues of Waterfall model? Explain.
10. Differentiate between functional and non-functional requirements. Discuss about the importance of a SRS document.
11. Write short notes on Any TWO:
(a) Structured analysis
(b) Software evolution
(c) Fault tolerance

2×3=6

